

Introduction to Information Retrieval

Landmark Retrieval

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Disclaimer

- Disclaimer: This curriculum have been inspired and adapted from materials created by:
 - Richard Szeliski, “Computer Vision: Algorithms and Applications”, 2nd edition, 2022. <https://szeliski.org/Book/>

Image Retrieval System



Google Landmarks Dataset v2

- GLDv2: 5M images and 200k distinct instance labels.
- Test set: 118k images with ground truth annotations for both the retrieval and recognition tasks

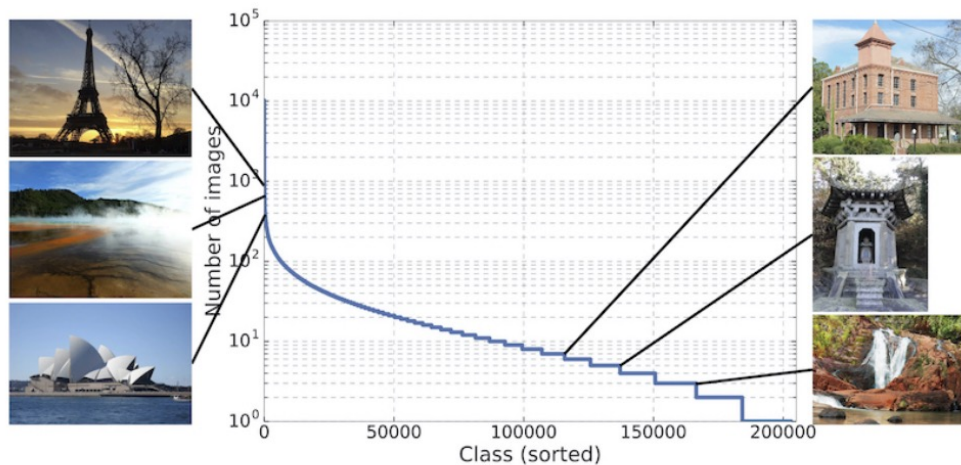


Figure 1: The Google Landmarks Dataset v2 contains a variety of natural and human-made landmarks from around the world. Since the class distribution is very long-tailed, the dataset contains a large number of lesser-known local landmarks.¹

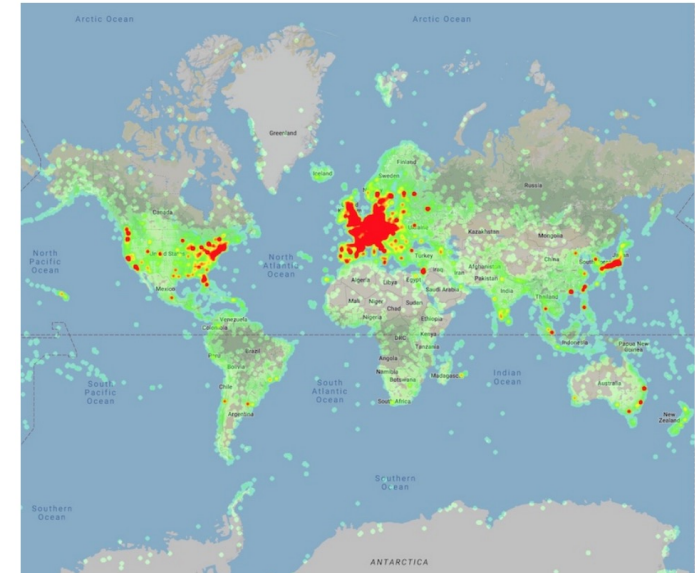


Figure 2: Heatmap of the places in the Google Landmarks Dataset v2.

Example

Query:



Relevant index images:



(b) Trevi Fountain – Different viewpoints and lighting conditions.

Example

Query:

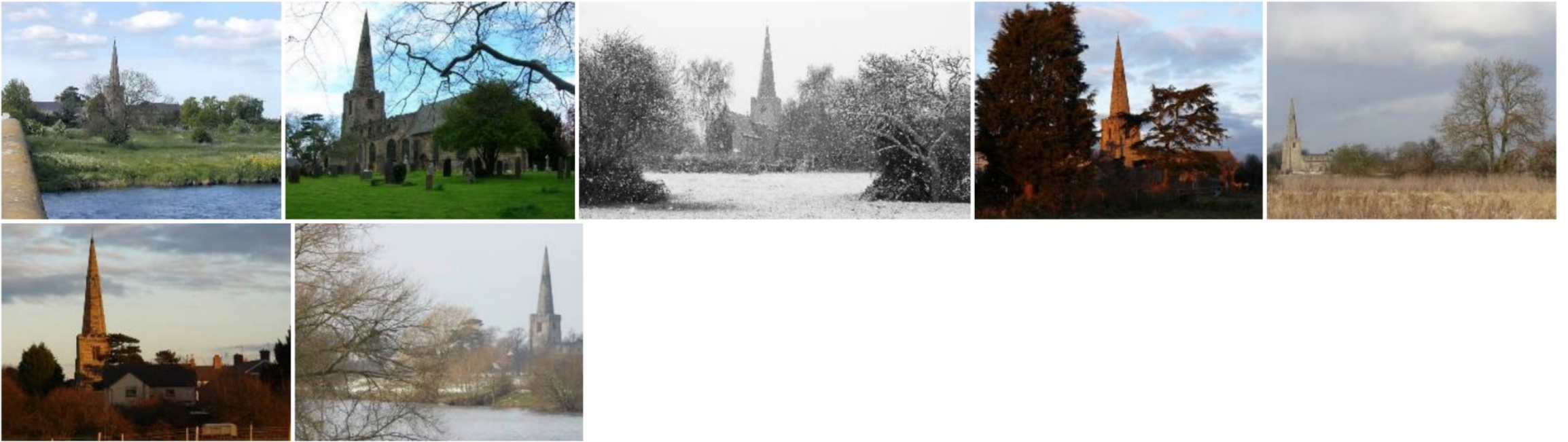


Relevant index images:



(b) Palazzo delle Esposizioni – Day and night photos and a monochrome print.

Example



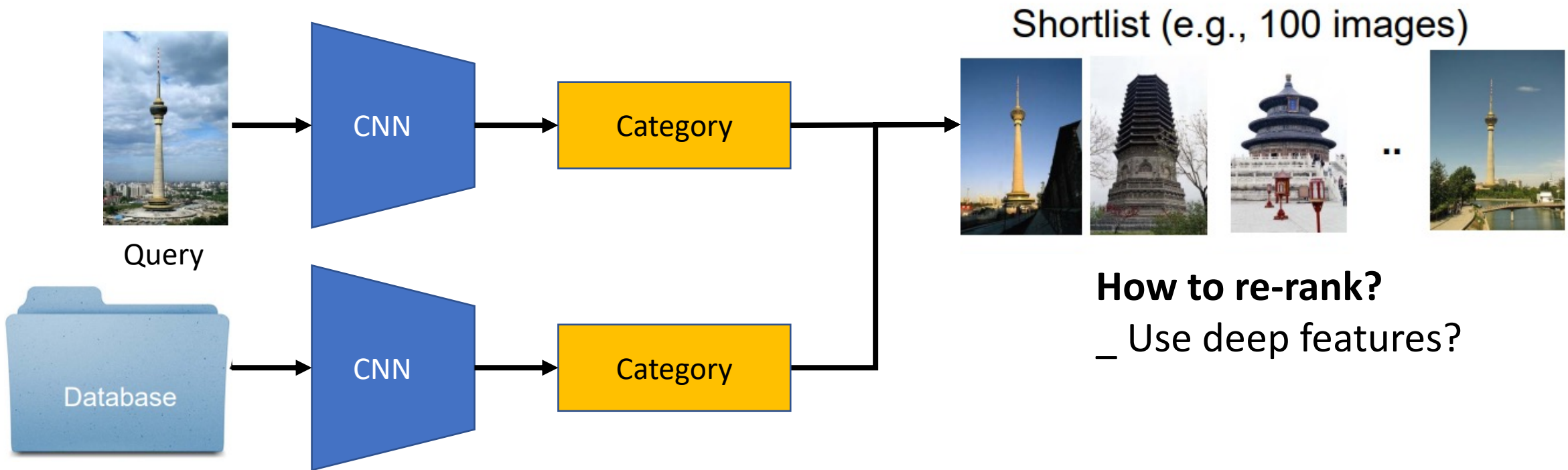
(c) All Saints church (Sawley) – Images showing the landmark in different seasons and under different lighting conditions.

Design Landmark Retrieval System?

- We have training set, in which all images in have labels
- How to use this information?

Design Landmark Retrieval System

- We have training set, in which all images in have labels
=> Train image classification network



Example

Query:



Relevant index images:



(a) Teatro Espanol – Some relevant images that are difficult to retrieve: Architectural drawing, painting of the inside, historical photograph of audience members.

Example



(d) Goryokaku (Hakodate) – Aerial views and details of a park.

Design Landmark Retrieval System

